

Week 5, Lecture 1 WOD: Answer Key

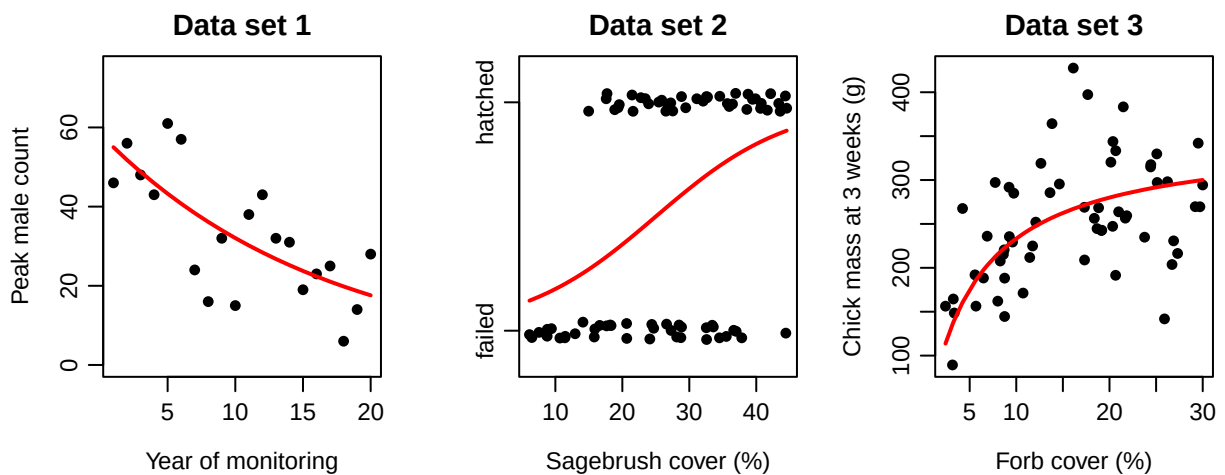
Choosing a Data-Generating Model

NRES 746, Instructor Answer Key (not for student distribution)

Each item repeats the handout's question, then gives the answer, a short rationale, and mistakes to watch for. Most items here have more than one defensible answer, so the Answer gives the model the data were simulated from (the "truth") plus acceptable variants. All numbers are computed in R when this file is knitted.

Whatever a student picks, look for: (1) a function whose shape matches the figure and respects the response's range; (2) a distribution that fits the data type (count, 0/1, positive continuous); (3) the noise parameter named, or a note that there isn't one; (4) two lines written as expected value plus noise around it.

```
# reproduces the handout's simulated data (same seed and order of draws)
set.seed(746)
yr <- 0:19
lek_mu <- 55 * exp(-0.06 * yr)
lek <- rnbinom(20, mu = lek_mu, size = 8)
cover <- runif(80, 5, 45)
p_true <- plogis(-2.5 + 0.1 * cover)
nest <- rbinom(80, 1, p_true)
forb <- runif(60, 2, 30)
mass_mu <- 350 * forb / (5 + forb)
mass <- rgamma(60, shape = 25, rate = 25 / mass_mu)
```



Red lines: the true expected values.

Part 1: Data set 1, lek counts

1a.

The plot shows the peak count at one lek over 20 years. Counts trend downward at roughly a

constant percentage per year, and the scatter around that trend is large. Expected value: ____
 Noise distribution: _____. Write your model.

Answer (truth). Negative exponential expected value, negative binomial noise:

$$\mu_t = N_0 e^{-rt} \quad Y_t \sim \text{NegBin}(\text{mean} = \mu_t, \text{size} = k)$$

(Simulated with $N_0 = 55$, $r = 0.06$, $k = 8$.)

Why. A constant *percentage* decline is what an exponential does: $\mu_{t+1}/\mu_t = e^{-r}$ every year. The noise parameter is k , with variance $\mu_t + \mu_t^2/k$.

Watch for.

- A straight line, $\mu_t = a - bt$, which eventually goes negative. Acceptable over the observed range if the student notes this.
- A Ricker, Michaelis-Menten, or logistic function (wrong shape for a steady decline).
- A negative binomial with no k .

1b.

Why this distribution?

Answer. Counts are discrete and non-negative, which points to Poisson or negative binomial. The scatter is too large for a Poisson, so negative binomial.

Why. A Poisson's variance equals its mean. These counts are far more spread out than that:

```
mean((lek - lek_mu)^2 / lek_mu)           # about 1 under a Poisson
```

```
## [1] 3.604747
```

```
mean(abs(lek - lek_mu) > 2 * sqrt(lek_mu)) # about 0.05 under a Poisson
```

```
## [1] 0.4
```

Watch for.

- Poisson with no comment on the spread. Poisson *with* a note that the scatter looks too big is a good answer.
- Normal with constant σ : defensible for counts this large, but its noise won't shrink as the mean falls.

Part 2: Data set 2, nest fate

2a.

Each nest either hatches at least one egg (success) or fails. Nests in denser cover look more likely to succeed. (The expected value of a 0/1 response is the probability of success, and a probability can never go below 0 or above 1, however much cover a nest has.) Write your model.

Answer (truth). Logistic expected value, Bernoulli noise:

$$p_i = \frac{1}{1 + e^{-(a+b \text{cover}_i)}} \quad Y_i \sim \text{Bernoulli}(p_i)$$

(Simulated with $a = -2.5$, $b = 0.1$.)

Why. The logistic rises with cover but stays between 0 and 1, as a probability must. There's no separate noise parameter: the variance, $p_i(1 - p_i)$, is set by the mean.

Watch for.

- A straight line with no comment on the bounds. It fits the observed range but leaves $[0, 1]$ when extrapolated:

```
predict(lm(nest ~ cover), data.frame(cover = c(0, 60)))
```

```
##           1           2
## -0.03024553  1.18902509
```

- Adding a σ to a Bernoulli model.
- Applying the function to the 0/1 outcome itself instead of to p_i .
- Acceptable variant: Michaelis-Menten with asymptote at most 1 (bounded and increasing).

2b.

Why this distribution?

Answer. The response takes only two values, 0 or 1. The Bernoulli is the distribution for a single success/failure trial.

Why. A normal, gamma, or Poisson can't put all its probability on 0 and 1.

Watch for. A binomial with $N = 80$. The nests are 80 separate Bernoulli trials, not one experiment with 80 trials. (Binomial with size 1 is fine.)

Part 3: Data set 3, chick mass

3a.

Mass rises with forb cover, but the gains shrink at high cover. Mass is always positive. Write your model.

Answer (truth). Michaelis-Menten expected value, gamma noise:

$$\mu_i = \frac{a \text{ forb}_i}{b + \text{forb}_i} \quad Y_i \sim \text{Gamma}(\text{mean} = \mu_i, \text{shape} = \alpha)$$

(Simulated with $a = 350$ g, $b = 5\%$, $\alpha = 25$.)

Why. Michaelis-Menten rises and levels off (“gains shrink”): a is the asymptotic mass and b the cover at half of a . Any saturating function in the grab bag is acceptable (e.g. a power law with exponent below 1).

Watch for.

- A Ricker (hump) or a straight line: wrong shape.
- Worth raising: Michaelis-Menten predicts zero mass at zero forb cover, which isn't biologically realistic, though it doesn't matter over the observed range.

3b.

Why this distribution?

Answer. Mass is continuous and positive, and its spread grows with the mean. A gamma with fixed shape has exactly that property.

Why. Residual spread rises with the mean while the relative spread (coefficient of variation) stays roughly constant (the truth is $1/\sqrt{25} = 0.2$):

```
bins <- cut(forb, c(0, 8, 16, 30))
res <- mass - mass_mu
rbind(sd = tapply(res, bins, sd), cv = tapply(res / mass_mu, bins, sd))
```

```
##           (0,8]      (8,16]      (16,30]
## sd 47.6218274 49.3848662 64.2053012
## cv  0.2933069  0.2072792  0.2281783
```

Watch for.

- Lognormal: also acceptable (positive, spread proportional to mean).
- Normal with constant σ : defensible here, since masses are far from zero, but ask whether the spread grows with the mean.
- Poisson: mass is continuous, not a count.

3c (if time).

If your distribution's own parameters are not the mean, work out how to express them using μ_i . (For a gamma with shape α and scale s , the mean is αs . What scale gives a mean of μ_i , and what does the shape do to the noise?)

Answer. $s = \mu_i / \alpha$. Then $\text{Var}(Y_i) = \alpha s^2 = \mu_i^2 / \alpha$, so the coefficient of variation is $1/\sqrt{\alpha}$: a larger shape means less relative noise.

Why. Set $\alpha s = \mu_i$ and solve. Checking by simulation:

```
alpha <- 25; mu <- 200
y <- rgamma(1e5, shape = alpha, scale = mu / alpha)
c(mean = mean(y), cv = sd(y) / mean(y), one_over_sqrt_alpha = 1 / sqrt(alpha))
```

```
##                mean                cv one_over_sqrt_alpha
##          200.1919685          0.1997803          0.2000000
```

Watch for. Confusing scale and rate (R's `rgamma()` takes either; rate = $1/\text{scale} = \alpha/\mu_i$). Bolker writes the shape as α ; accept any consistent notation.

Part 4: If time, a fourth data set

4.

Telemetry crews record how far each radio-collared hen moves between consecutive daily locations (meters). Most daily moves are short (a typical hen moves 100-200 m), but a few days involve moves of more than a kilometer. There is no predictor in this data set. What would you use for the expected value, and which distribution for the noise? Does the expected value need to depend on a predictor at all? Explain.

Answer. A **constant** expected value, $\mu_i = \mu$, with a positive, right-skewed distribution: exponential, gamma, or lognormal.

Why. With no predictor, the expected value is just a constant (an intercept-only model is still a model). Distances are positive and a few are very long, so the distribution needs a long right tail. With a mean near 250 m, all three candidates match the description:

```
mu <- 250
cv <- 1.5; s_ln <- sqrt(log(1 + cv^2)); m_ln <- log(mu) - s_ln^2 / 2
data.frame(
  dist = c("exponential", "gamma, shape 0.8", "lognormal, CV 1.5"),
  median_m = round(c(qexp(0.5, 1 / mu), qgamma(0.5, 0.8, scale = mu / 0.8), exp(m_ln))),
  p_over_1km = round(c(pexp(1000, 1 / mu, lower.tail = FALSE),
    pgamma(1000, 0.8, scale = mu / 0.8, lower.tail = FALSE),
    plnorm(1000, m_ln, s_ln, lower.tail = FALSE)), 3))
```

```
##                dist median_m p_over_1km
## 1      exponential      173      0.018
## 2   gamma, shape 0.8      157      0.026
## 3 lognormal, CV 1.5      139      0.034
```

Watch for. A normal (symmetric, allows negative distances, no long tail); a Poisson (distance is continuous); “no predictor, so no model.”